

Task Management System

A Lightweight Productivity Web Application

Group: TaskFlow

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Problem Statement

- Difficulty in managing daily tasks efficiently
- Manual tracking leads to missed deadlines
- Existing platforms are feature-heavy or complex
- Need for a simple and lightweight solution

Proposed Solution

- Web-based Task Management System
- Secure user authentication
- Create, update, delete tasks (CRUD)
- Priority-based categorization
- Simple and user-friendly interface

Technology Stack

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** FastAPI (Python-based framework)
- **Database:** SQLite3
- **Security:** Password Hashing (Passlib/Bcrypt)
- **Configuration:** Environment Variables (.env)

System Architecture

Frontend

HTML, CSS, JavaScript – User Interface

Backend

FastAPI – API handling, authentication, logic

Database

SQLite3 – Task & User data storage

Security Implementation

- Secure user authentication system
- Password hashing before database storage
- Environment variables for secret configuration
- Database file excluded from version control
- Input validation at backend layer

Comparison with Existing Systems

Feature	Trello/Asana	Notion	TaskFlow
Complexity	High	Moderate	Low
Customization	High	High	Basic
Ease of Use	Moderate	Moderate	High
Lightweight Design	No	No	Yes
Learning Purpose	No	No	Yes

Project Timeline (1 Month Plan)

- **Week 1:** Requirement Analysis and System Design
- **Week 2:** Frontend Development (UI Implementation)
- **Week 3:** Backend Development and Database Integration
- **Week 4:** Testing, Security Implementation, and Deployment Preparation

Current Status of Project

- Project idea finalized
- Requirements gathered and analyzed
- Tech stack selected
- System architecture designed
- Development phase to begin

- Cloud deployment (Render/Railway) for remote access
- Role-based user access control
- Real-time task updates
- Mobile responsive interface
- Integration with AI-based task prioritization